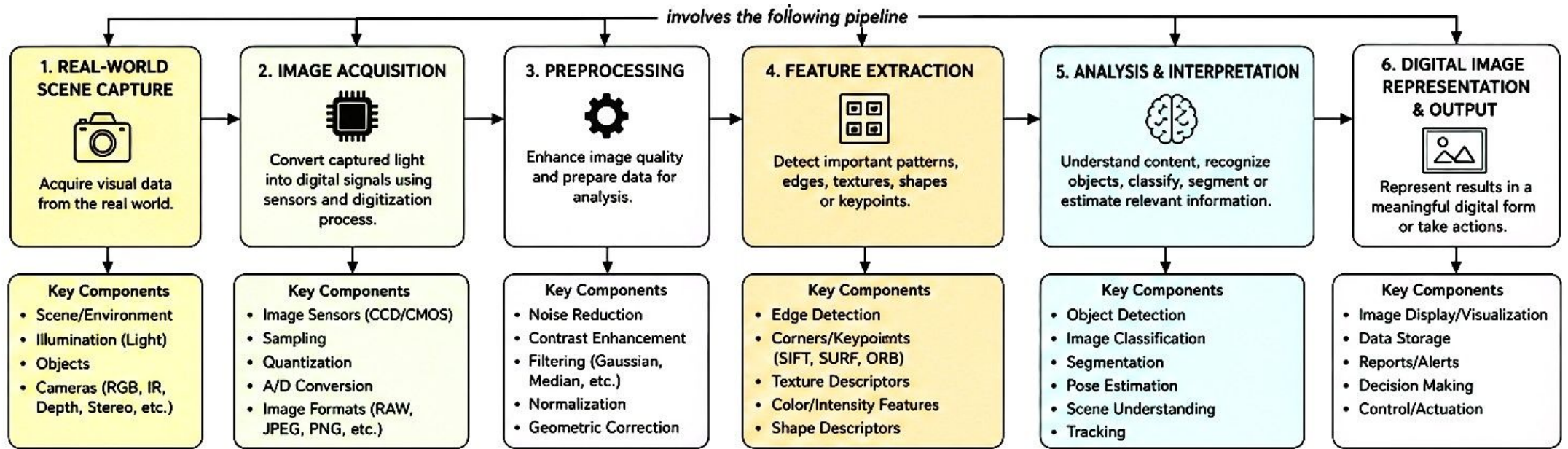


Core Capabilities of CV Systems



Selection of Appropriate Vision-Based Applications

- Based on problem requirements (accuracy, speed, real-time, scale).
- Consider data availability, environment, cost and deployment needs.
- Match system capabilities with application objectives.

Influence on Application Effectiveness

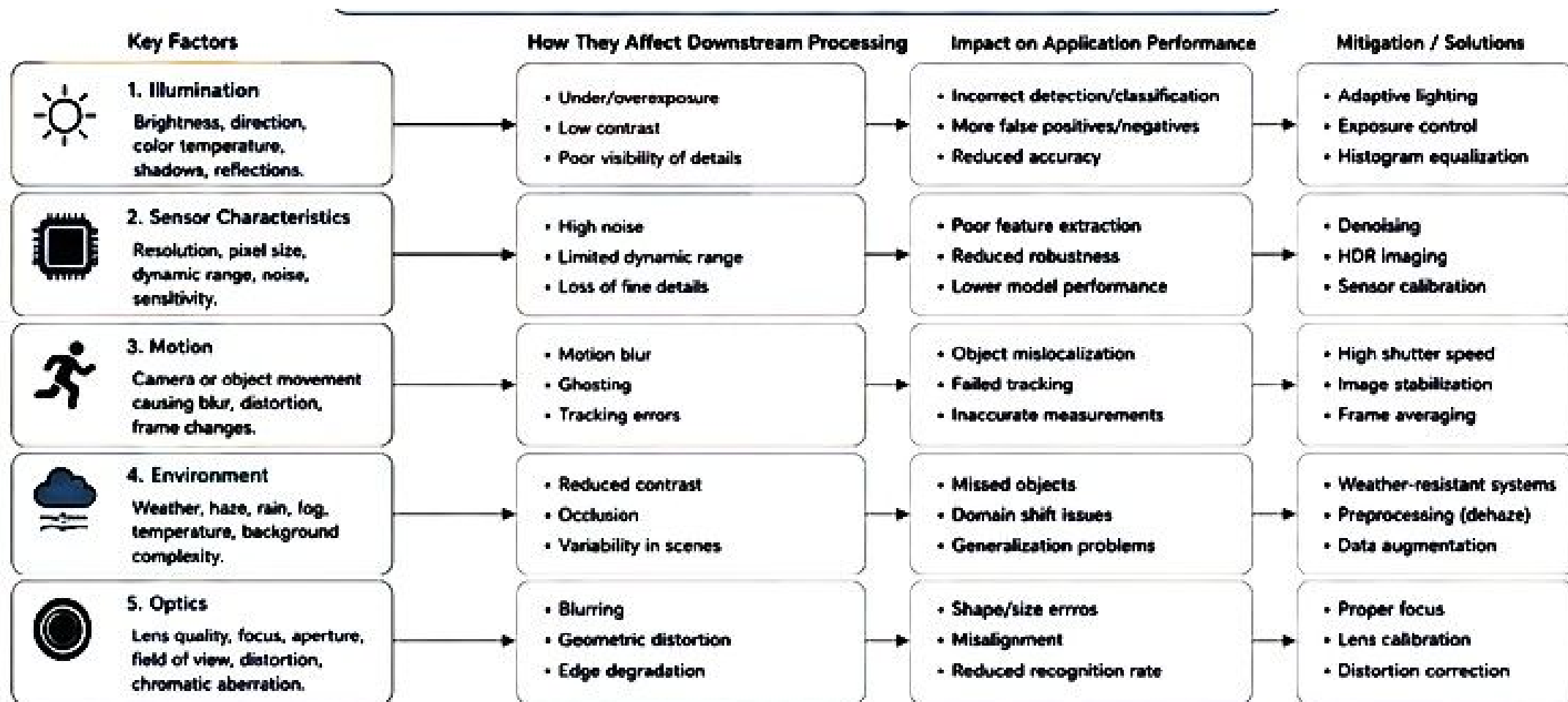
- Better capabilities → Higher accuracy, reliability and robustness.
- Proper application choice → Efficient resource usage and ROI.
- Real-time performance, scalability and user satisfaction improved.

Integration of Perspectives from Multiple Literature Sources

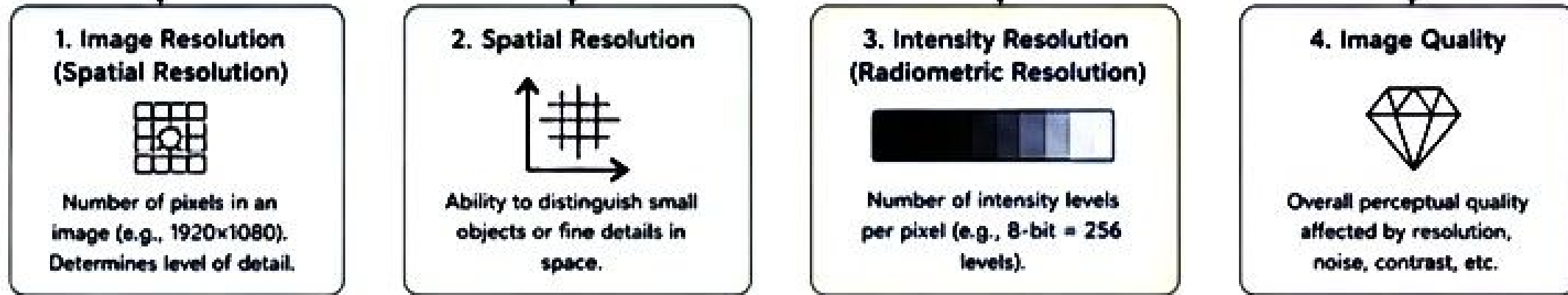


Outcomes





Types of Resolution



Cause-Effect Relationships



Impact on Image Usability

- Higher resolution → More detail → Better object detection & classification.
- Low resolution → Loss of detail → Missed/incorrect detections.
- Very high resolution → Large storage & computation cost.

Sources of Evidence (Examples)

- Gonzalez & Woods, Digital Image Processing
- Szeliski, Computer Vision: Algorithms & Applications
- Recent research papers in IEEE, Springer, Elsevier

Applications



